

FAST– National University of Computer and Emerging Sciences

DS2001 Introduction to Data Science

Fall 2025

Assignment#1 (Deadline: 22th September 2025)

- Submit your Jupyter Notebook (.ipynb) file and summary report (.pdf or .docx).
- This Assignment will be completed in group of 2 students. The goal is to collaboratively clean, analyze, and explore the

TASK # 1(A): (5marks)

Identify where interesting datasets relevant to the following domains can be found on the internet and explain what you must do to turn this data into a usable format on your computer for analysis.

1. Books
2. Horse Racing
3. Risks and Diseases
4. Crime rates
5. Bird watching

TASK # 1B: (5marks)

Visit <http://data.gov>, and Identify five datasets that sound interesting to you. For each write and propose three interesting things you might do with them.

TASK # 2-Data Cleaning and EDA: (20 marks)

Load Titanic dataset from seaborn. Apply data cleaning techniques: handling missing values, duplicates, formatting, and outliers. Perform Exploratory Data Analysis (EDA). After cleaning and basic EDA, answer the following:

IRIS-Dataset

1. Which flower species has the largest average petal length?
2. Do sepal length and petal length appear correlated? If yes, is it stronger for some species than others?
3. Which feature has the least overlap between species (best for classification)?
4. Are there any outliers in petal width? If so, which species do they belong to?

5. If you had to classify a new flower using only one feature, which one would you choose and why?
6. Do flowers with shorter sepals tend to have narrower petals?

Titanic Dataset

- 1) What percentage of males vs females survived?
- 2) Was class (1st, 2nd, 3rd) a stronger survival factor than gender? Support with numbers/plots.
- 3) What is the average age of survivors vs non-survivors?
- 4) Do children (age < 12) have a higher survival rate compared to adults?
- 5) Among passengers traveling alone vs with family (based on sibsp and parch), who had a better survival rate?
- 6) Is there any evidence that fare influenced survival probability? (Hint: plot boxplot of fare vs survival).
- 7) Does the embarkation port (C, Q, S) show differences in survival rate?
- 8) Which group had the lowest survival probability? (gender + class + age).
- 9) Do survival chances correlate more with socioeconomic class (fare/class) or with biological factors (gender/age)?

TASK # 3-Data Cleaning and EDA: (10 marks)

Provide answers to the questions associated with the following data sets, available at <http://www.data-manual.com/data>

- a) Analyze the movie data set. What is the range of movie gross in the United States? Which type of movies are most likely to succeed in the market? Comedy? PG-13? Drama?
- b) Analyze the Manhattan rolling Sales dataset. Where in Manhattan is the most/least expensive real estate located? What is the relationship between Sales Price and gross square feet?
- c) Analyze the 2012 Olympic dataset. What can you say about the relationship between a country's population and the number of medal it wins? What can you say about the relationship between the ratio of female and male counts and the GDP of that country?
- d) Analyze the GDP per capita dataset. How do countries from Europe, Asia and Africa compare in the rates of growth in GDP? When have countries faced substantial changes in GDP, and what historical events were likely most responsible for it?